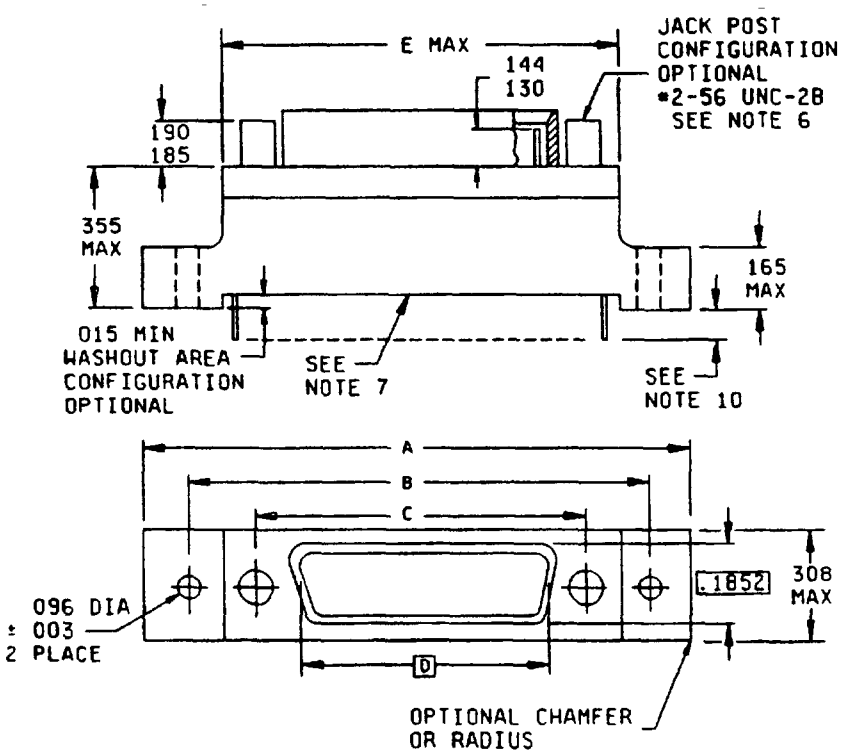
INCH-POUND
MIL-C-83513/25A
14 Oct 1993
SUPERSEDING
MIL-C-83513/25
11 February 1992

MILITATRY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, STRAIGHT, SOCKET CONTACTS, 2 ROW ,SOLDER TYPE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513.

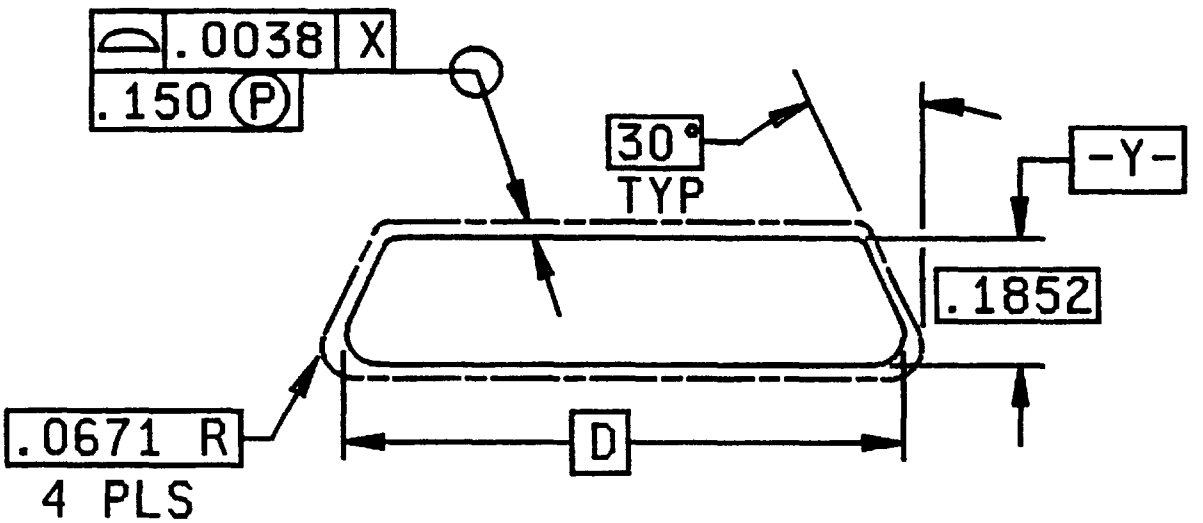


Inches	mm	Inches	mm	Inches	mm
.003	0.08	130	3.30	1852	4.704
0038	0 097	.144	3 66	.190	4.83
.015	0.38	.150	3 81	.308	7.83
.0671	1.704	.165	4 19	355	9 02
096	2 44	.185	4 70		

FIGURE 1 Connector, receptacle .050 spacing

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DETAIL OF INTERFACE

FIGURE 1 Connector, receptacle, 050 spacing - Continued

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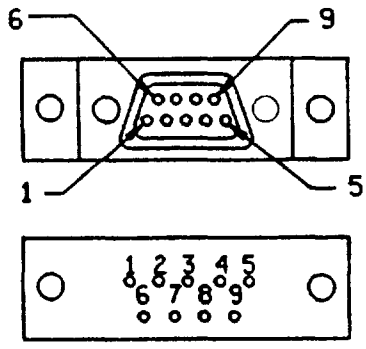
Number of contacts	A Max	B ±.007 (0.18 mm)	C ±.005 (0.13)	D	E Max
9	1.390 (35.31)	1.150 (29.21)	.565 (14.35)	.3342 (8.489)	.885 (2.48)
15	1.390 (35.31)	1 150 (29.21)	.715 (18.16)	.4842 (12.299)	.945 (24.00)
21	1.690 (40.64)	1.450 (36.83)	.865 (21.97)	.6342 (16 109)	1.185 (30.10)
25	1.740 (44.20)	1.500 (38 10)	.965 (24.51)	.7342 (18.649)	1.275 (32.39)
31	2.040 (51.82)	1.800 (45.72)	1.115 (28.32)	.8842 (22.459)	1.575 (40.01)
37	2.340 (59.44)	2.100 (53.34)	1.265 (32.13)	1 0342 (26 269)	1.875 (47.63)

NOTE: Metric equivalents are in parentheses

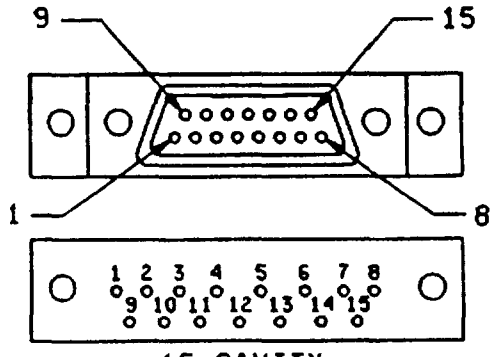
NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± 005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513 Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
5. Metal shell shall be of material in accordance with MIL-C-83513 for class M parts.
6. Jackpost (permanently attached), when specified: Corrosion-resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.
7. Separately molded plastic body (if used) shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
8. Wire termination sockets shall conform to QQ-W-343, no. 24 AWG copper.
9. Interfacial seals are included with each type receptacle connector.
10. Termination lengths available: .109 (2.77 mm), 140 (3 56 mm), or .172 (4 37 mm). The tolerance shall be ± 015 (0 38 mm) for all termination lengths

FIGURE 1 Connector, receptacle, 050 spacing - Continued

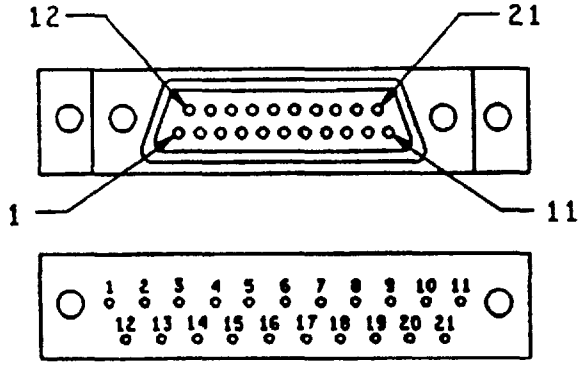
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9 CAVITY
TERMINATION VIEW - VERTICAL MOUNT



15 CAVITY
TERMINATION VIEW - VERTICAL MOUNT

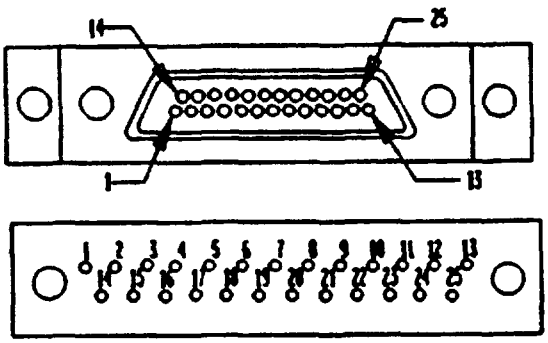


21 CAVITY
TERMINATION VIEW - VERTICAL MOUNT

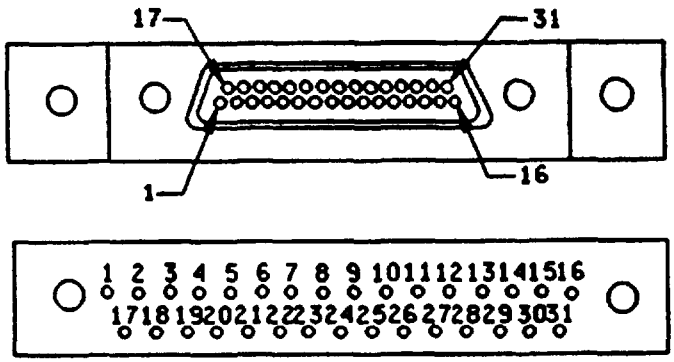
NOTE Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part

FIGURE 2 Insert arrangement.

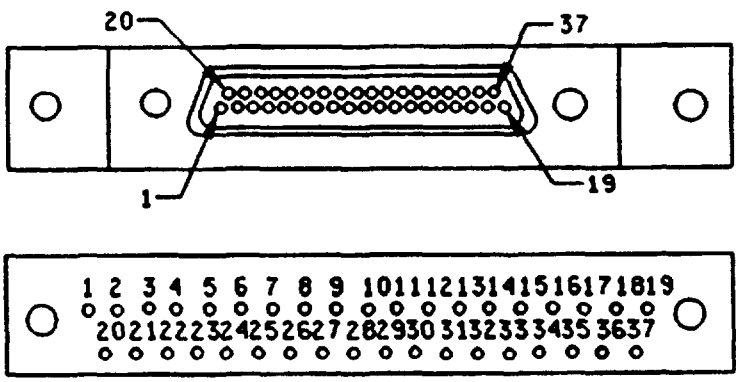
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25 CAVITY
TERMINATION VIEW - VERTICAL MOUNT



31 CAVITY
TERMINATION VIEW - VERTICAL MOUNT

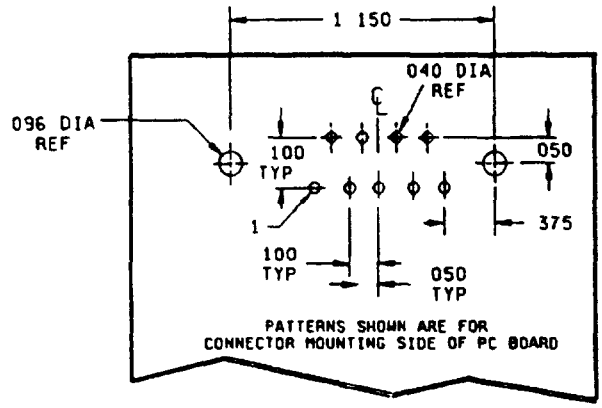


37 CAVITY
TERMINATION VIEW - VERTICAL MOUNT

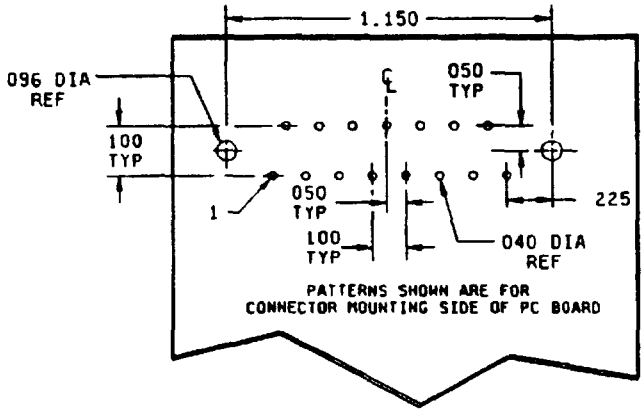
NOTE: Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part

FIGURE 2 Insert arrangement - Continued

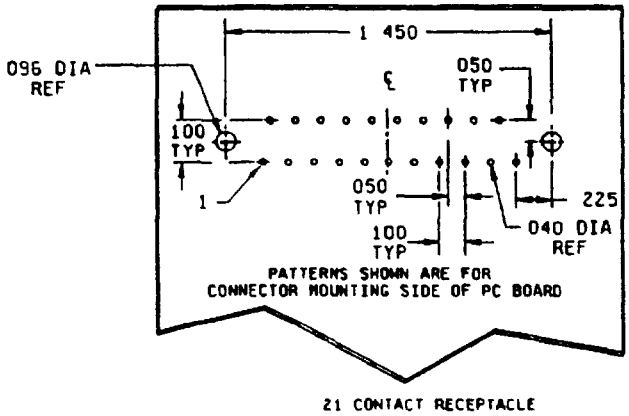
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9 CONTACT RECEPTACLE



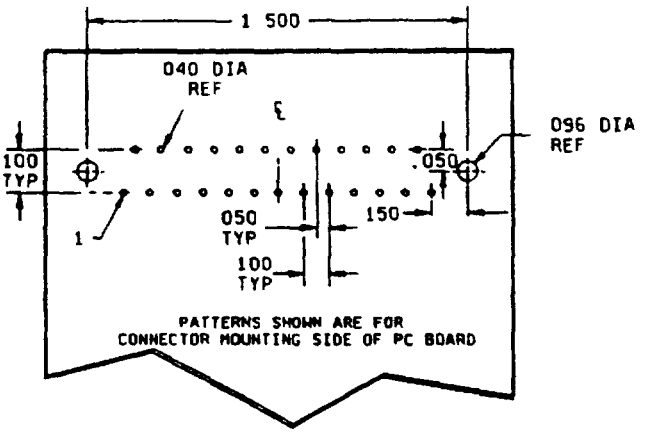
15 CONTACT RECEPTACLE



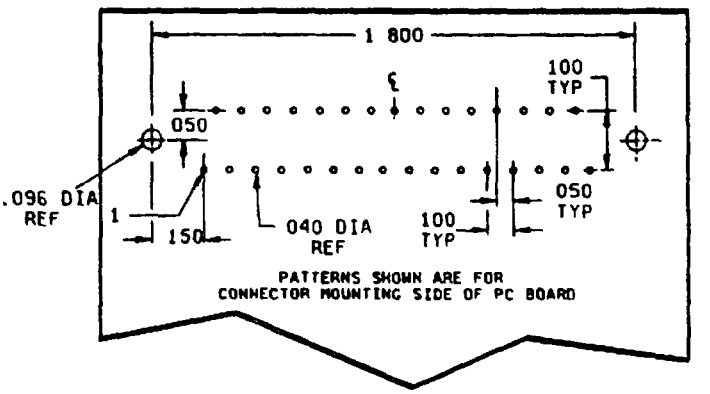
21 CONTACT RECEPTACLE

FIGURE 3 Layout arrangement

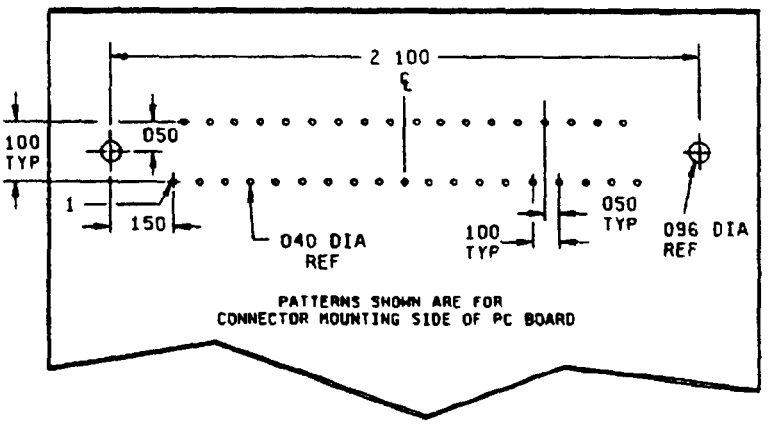
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25 CONTACT RECEPTACLE



31 CONTACT RECEPTACLE



37 CONTACT RECEPTACLE

Inches	mm
.040	1.02
.050	1.27
.096	2.44
.100	2.54
.150	3.81
.225	5.72
.375	9.53
1.150	29.21
1.450	36.83
1.500	38.10
1.800	45.72
2.100	53.34

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for infomation only

FIGURE 3 Layout arrangement - Continued

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REQUIREMENTS

Dimensions and configurations See figures 1 and 2.

Current rating, maximum 3 amperes per contact

Materials

Termination organization area. Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513

Shell The requirements for shell materials shall be in accordance with MIL-C-83513

Plastic body or plastic molding. Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.

Jackpost. Corrosion-resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.

Wire termination pins. Wire termination pins shall conform to QQ-W-343, no 24 AWG copper.

Mating connector: Shall conform to MIL-C-83513/1 and MIL-C-83513/3.

Plating of termination leads: Solder dipping socket of termination leads will be accomplished in SN60 PB40 or SN63 PB37 in accordance with QQ-S-571

Part or Identifying Number (PIN) Consists of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code for the shell finish and the hardware

M83513/25-	D	01	C	P
Specification sheet number	Insert arrangements (see figure 2) A = 09 B = 15 C = 21 D = 25 E = 31 F = 37	Termination length 01 = 109 02 = 140 03 = 172	Shell finish C = Cadmium N = Electroless nickel (space application)	Hardware N = No jackpost P = Jackpost attached

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities
Army - MI
Navy - AS, SH
Air Force - 17, 99
DLA - ES

User activities
Army - AT, ME
Navy - CG, MC

Preparing activity.
Air Force - 85

Agent
DLA - ES

(Project 5935-3921-16)

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